CSE 270 Sports Analytics

Greatest of All Time (GOAT) Analysis

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Libraries used

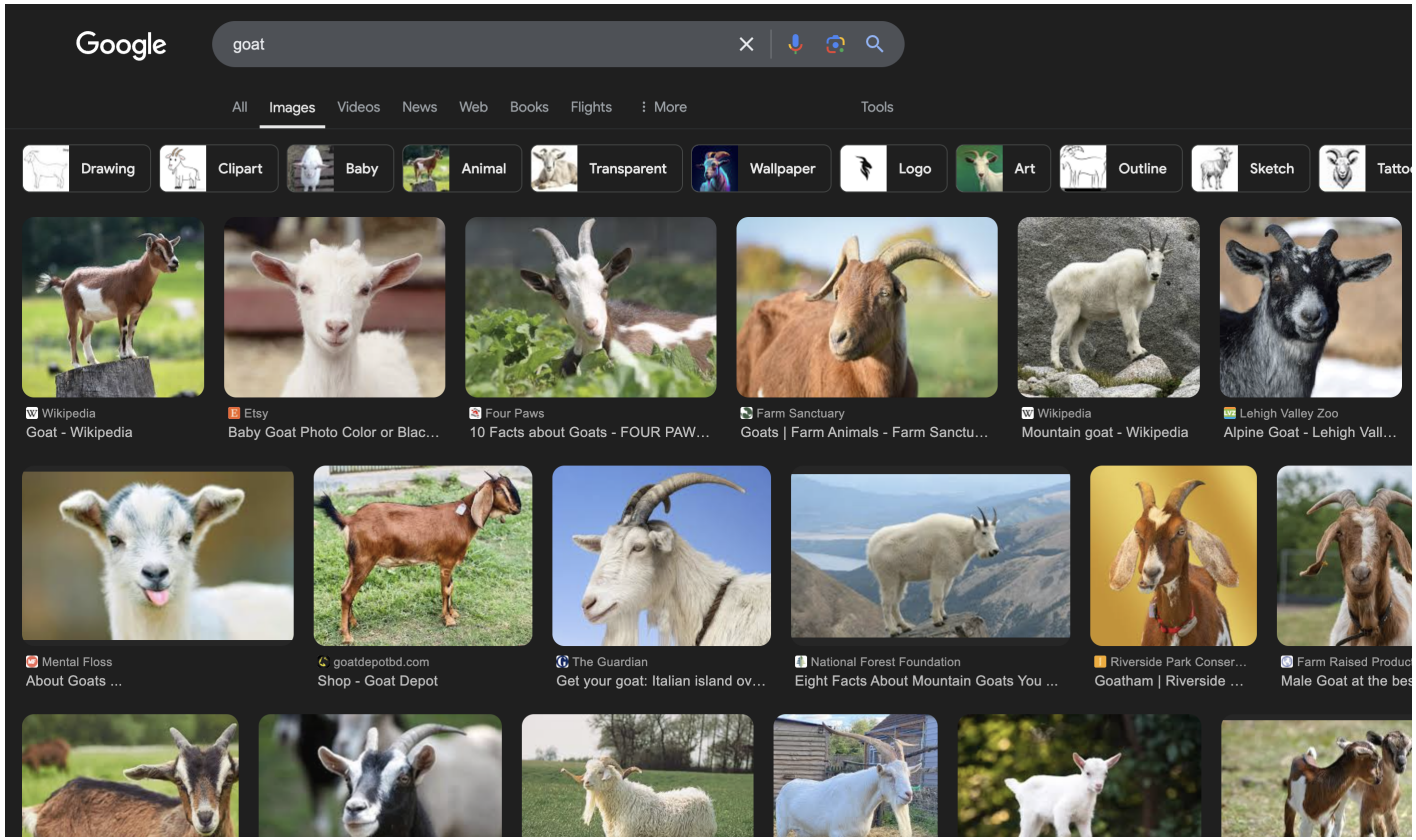
```
library(dplyr)
library(ggplot2)
library(gridExtra)
library(Lahman)
library(tidyr)
```

Introduction

What is GOAT ?

GOAT

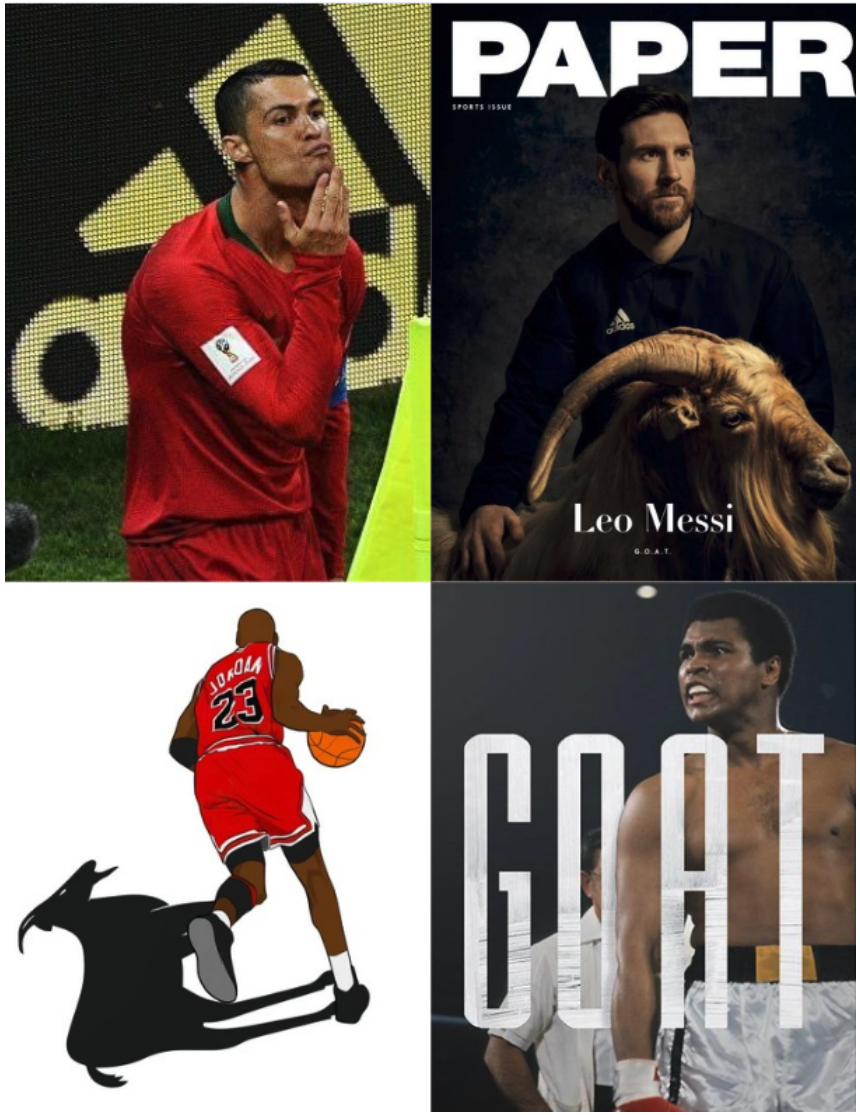
Results according to google.com:



GOAT

- GOAT(Greatest of All Time) is a term used to describe the greatest player in a given sport.
- The GOAT debate is one of the most passionate converstaion starters in sports.
- The concept of worshiping athletes came up from the ancient Greek mythology.
- The GOAT term was popularized in the 1990s in the hip-hop culture.
- The Social media has made the GOAT debate more popular.

GOAT



GOAT

Who are the GOATs in different sports?

GOAT

This is a question that has no official answer and is open to interpretation.

Sources of studies for GOAT analysis:

- Marca article
- Forbes article

GOAT

How to find the “GOAT” of each sport ?

GOAT

- Asking the local barber ?
- Conducting a survey ?
- Analyzing the players' statistics ?

GOAT

- What are the main characteristics of a good sports player?
- What are the main characteristics of a GOAT?
- How to measure the greatness of a sports player?
- Is the “GOAT”ness adaptable to each sport, or each player has a different “GOAT”ness level ?

GOAT

For some sports the "GOAT"ness of the player is more obvious than for others.
Whereas for some other sports there is an ongoing debate.

- Michael Jordan in basketball
- There is even a saying "Michael Jordan of something" to describe the best in a field.
- Zinedine Zidane in football(I just think so !)
- Wayne Gretzky in ice hockey.
    - He has the most goals and assists in the history of the NHL.

GOAT

- The GOAT players are expected to have rare abilities and skills and domination over the others.
- The GOAT type of players are pretty rare in sports history.
- In the language of statistics we need to refer back to the distributions.
- The GOAT players are expected to be in the right tail of the distribution of the players.

GOAT

We are going to cover 3 different sports and identify the GOAT from each of the sports.

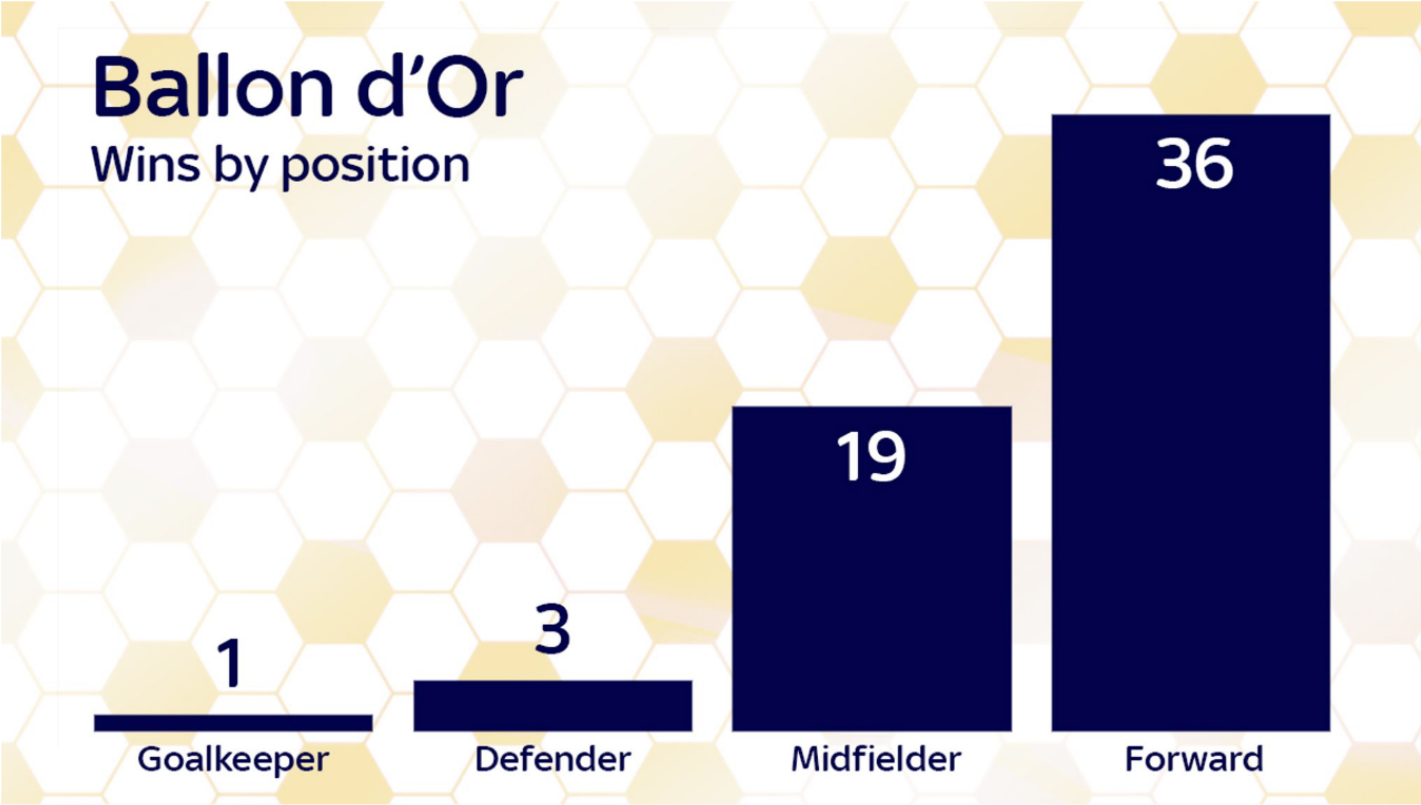
Let's start with FOOTBALL !

GOAT

What were the main GOAT debates over the years ?

- Maradona vs Pele
  - Pele has won 3 world cups, Maradona has won 1.
  - Maradona has major trophies on the club level as well.
  - None of them has a single Ballon D'or.
- Messi vs Ronaldo
  - Messi has more Ballon D'ors, Ronaldo has more Champions League trophies.
  - Messi shined only for Barcelona and Argentina, Ronaldo did it everywhere.
  - Ronaldo is more complete player with a wider range of skills.
  - Messi is the best in every skill he has.
  - Messi won every possible trophy.

GOAT


Ballon d'Or
Wins by position
1
3
19
36
Goalkeeper
Defender
Midfielder
Forward

GOAT

In football the most prestigous individual award is the Ballon D'or. It is awarded to the best player in the world every year.

- The attacking players and midfielders have the highest number of awards.
- Only one goalkeeper and 4 central defenders won the trophy and no full back has ever won it.
- Players like Maradona, Pele, Alfredo Di Stefano, got special ballon d'or awards after their career.
- Pele is on the top of the lost for the player of the century. So we will consider him as the "GOAT" by the end of the 20th century.
- Now let's find the GOAT of the 21st century.

GOAT

How to evaulaute an attacking players ?

GOAT

We will look for the seasonal statistics of the players who played in the top 5 leagues and in the Champions league and Europa League.

```
load("./data/tm_players_data.rda")
str(tm_players_data)
```

```
## 'data.frame':    36040 obs. of  18 variables:
##  $ Season           : int  2006 2004 2004 2003 2002 2002 2001 2001 2000 2000 ...
##  $ Competition      : chr  "DFB-Pokal" "LaLiga" "Premier League" "Premier League"
##  $ Squad            : int  2 16 1 12 23 1 29 1 33 2 ...
##  $ Appearances      : num  2 16 1 11 23 1 29 1 33 2 ...
##  $ PPG              : num  1.5 1.44 0 0.55 1.57 0 0.79 3 1.21 3 ...
##  $ Goals            : num  1 1 0 0 2 0 6 1 6 0 ...
##  $ Assists          : num  0 0 0 0 2 0 1 0 2 0 ...
##  $ OwnGoals         : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ SubsOn           : num  0 1 1 4 6 1 1 0 0 0 ...
##  $ SubsOff          : num  0 7 0 3 5 0 2 0 6 0 ...
##  $ YellowCards      : num  0 1 0 1 3 0 2 0 4 0 ...
##  $ SecondYellowCards: num  0 0 0 0 0 0 0 0 0 0 ...
##  $ RedCards         : num  1 0 0 0 0 0 0 0 0 0 ...
##  $ PenaltyGoals     : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ MinutesPerGoal   : num  205 1.2 0 0 779 ...
##  $ MinutesPlayed    : num  205 1.21 25 617 1.56 ...
##  $ PlayerName       : chr  "Mustapha Hadji" "Mustapha Hadji" "Mustapha Hadji" "Mu
```

GOAT

Let's summarize the data for all seasons and for all players.

```
tm_players_data <- tm_players_data %>%
  group_by(PlayerName) %>%
  summarise(
    Apperances = sum(Appearances),
    Goals = sum(Goals),
    Assists = sum(Assists),
    PPG = mean(PPG),
    PenaltyGoals = sum(PenaltyGoals),
    GoalContributions = Goals + Assists,
    MinutesPerGoal = mean(MinutesPerGoal),
    MinutesPlayed = sum(MinutesPlayed),
    MinutesPerGc = MinutesPlayed/GoalContributions,
    GamesWon = as.integer((MinutesPlayed / 96) * PPG),
    GCPer90 = GoalContributions / (MinutesPlayed / 90)
  )
```

GOAT

Let's sort out only the players who played more than average in terms of Minutes Played on the field.

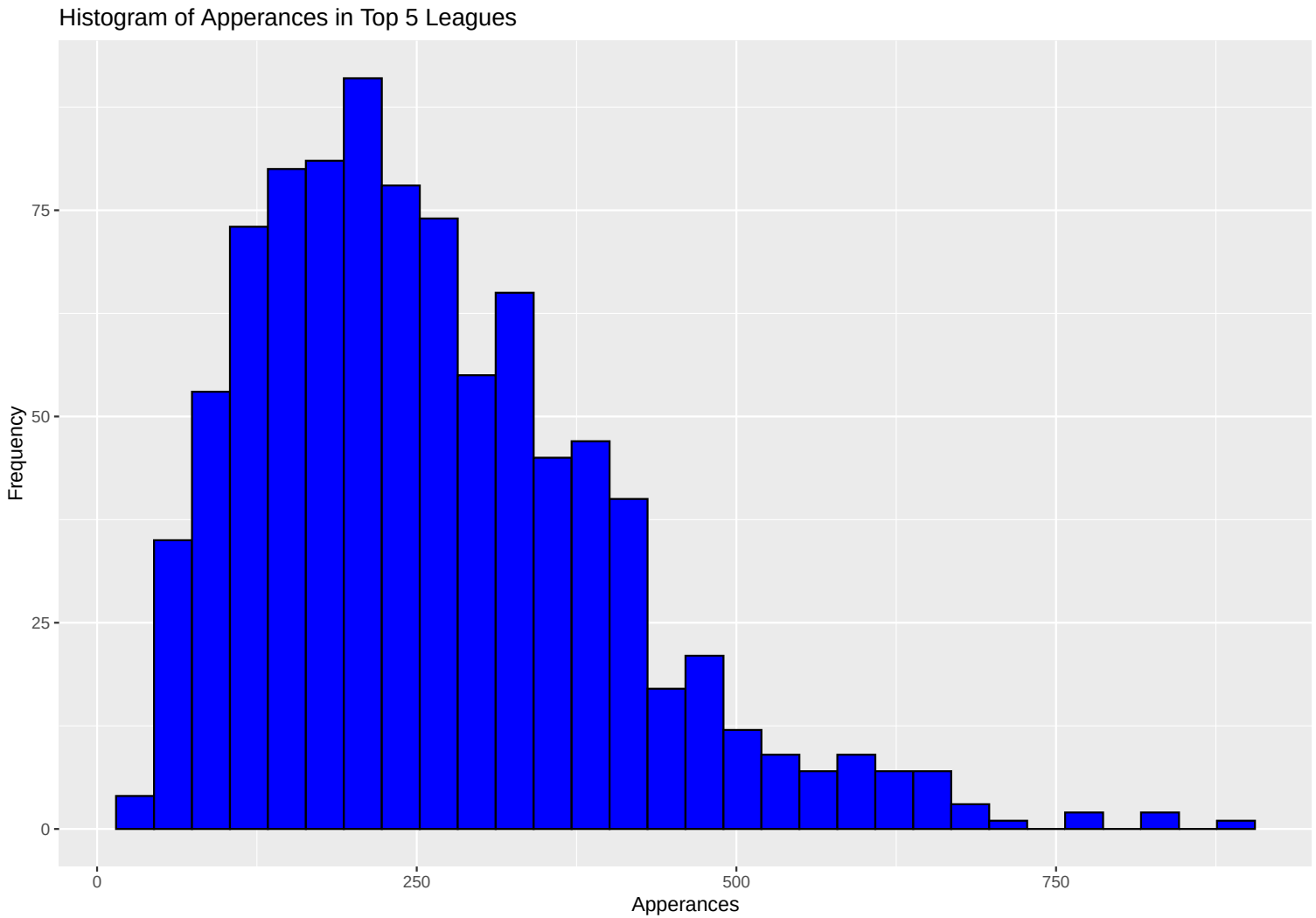
Let's analyze the distributions of all of the metrics.

```
tm_players_data <- tm_players_data %>%
  filter(MinutesPlayed > mean(MinutesPlayed))
```

GOAT

```
(apperances_histogram <- ggplot(data = tm_players_data, aes(x = Apperances)) +
  geom_histogram(bins = 30, fill = "blue", color = "black") +
  labs(title = "Histogram of Apperances in Top 5 Leagues",
       x = "Apperances", y = "Frequency"))
```

GOAT


Histogram of Apperances in Top 5 Leagues
Frequency
75
50
25
0
0
250
500
750
Apperances

GOAT

Top 5 players with the maximum number of Apperances.

```
tm_players_data %>%
  arrange(desc(Apperances)) %>%
    select(PlayerName,Apperances) %>%
      head(5)
```

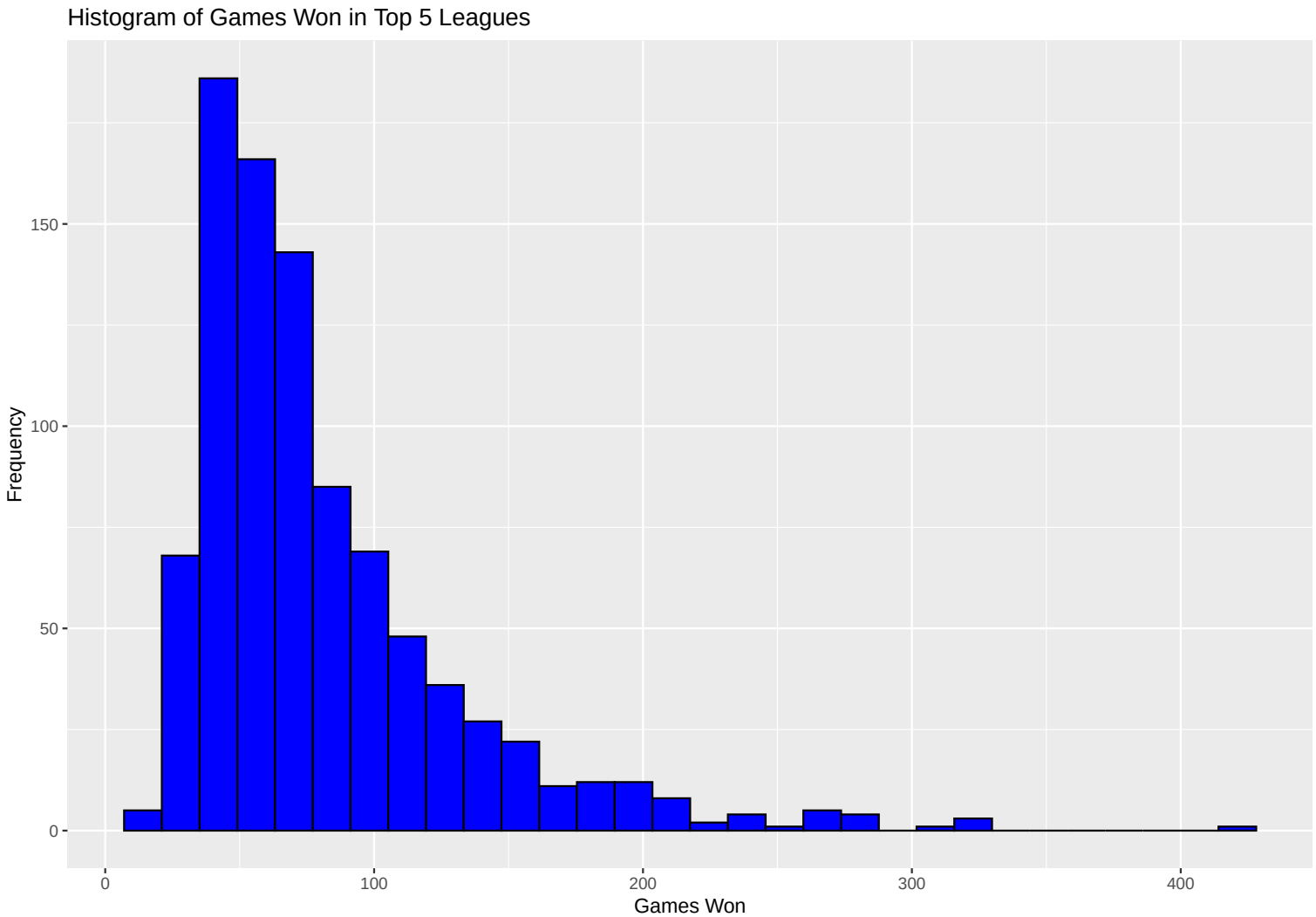
```
## # A tibble: 5 x 2
##   PlayerName        Apperances
##   <chr>                  <dbl>
## 1 Cristiano Ronaldo        893
## 2 Lionel Messi             843
## 3 Joaquin                  818
## 4 Raul Garcia              782
## 5 Karim Benzema            779
```

GOAT

Now let's identify the players who brought most of the wins to their teams when on the field.

```
(games_won_histogram <- ggplot(data = tm_players_data, aes(x = GamesWon)) +
  geom_histogram(bins = 30, fill = "blue", color = "black") +
  labs(title = "Histogram of Games Won in Top 5 Leagues",
       x = "Games Won", y = "Frequency"))
```

GOAT


Histogram of Games Won in Top 5 Leagues
Frequency
150
100
50
0
0
100
200
300
400
Games Won

GOAT

Top 5 players with the maximum number of games won.

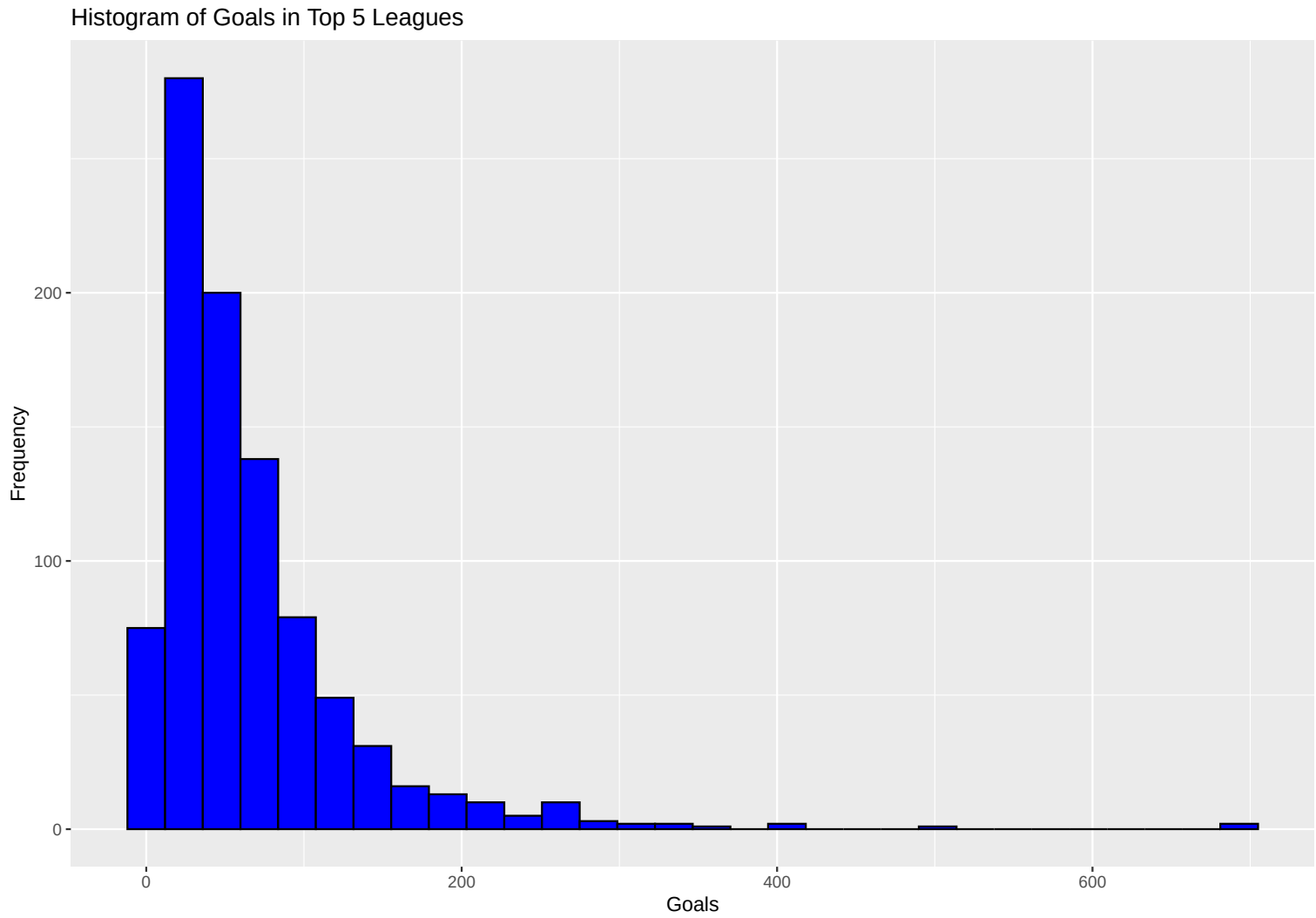
```
tm_players_data %>%
  arrange(desc(GamesWon)) %>%
    select(PlayerName,GamesWon) %>%
      head(5)
```

```
## # A tibble: 5 x 2
##   PlayerName          GamesWon
##   <chr>                  <int>
## 1 Lionel Messi             421
## 2 Robert Lewandowski       327
## 3 Karim Benzema            326
## 4 Angel Di Maria           320
## 5 Cristiano Ronaldo        307
```

GOAT

```
(goals_histogram <- ggplot(data = tm_players_data, aes(x = Goals)) +
  geom_histogram(bins = 30, fill = "blue", color = "black") +
  labs(title = "Histogram of Goals in Top 5 Leagues",
       x = "Goals", y = "Frequency"))
```

GOAT


Histogram of Goals in Top 5 Leagues
Frequency
200
100
0
0
200
400
600
Goals

GOAT

Top 5 goal scorers

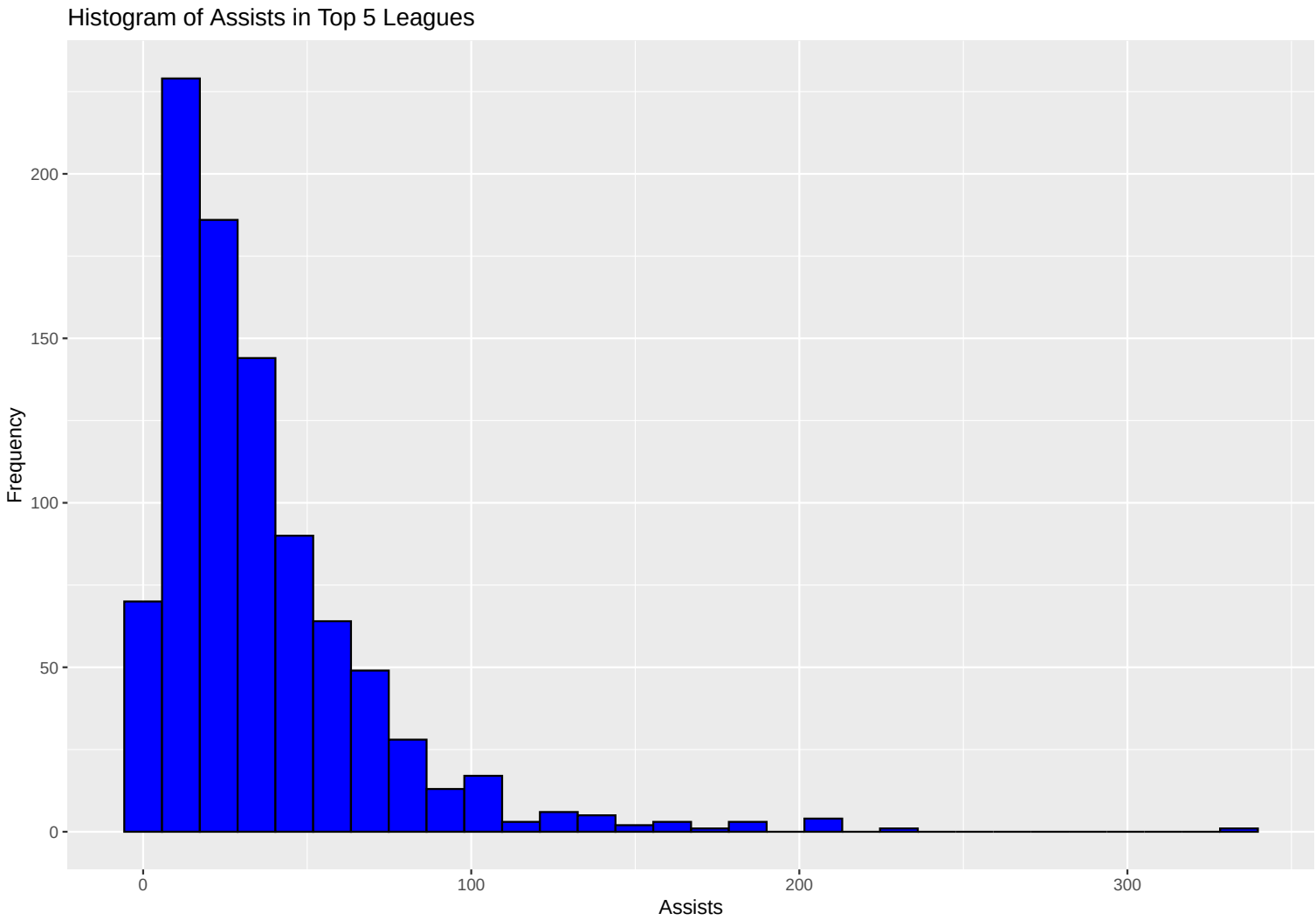
```
tm_players_data %>%
  arrange(desc(Goals)) %>%
    select(PlayerName,Goals) %>%
      head(5)
```

```
## # A tibble: 5 x 2
##   PlayerName         Goals
##   <chr>              <dbl>
## 1 Lionel Messi         695
## 2 Cristiano Ronaldo    682
## 3 Robert Lewandowski   504
## 4 Karim Benzema        413
## 5 Zlatan Ibrahimovic   395
```

GOAT

```
(assists_histogram <- ggplot(data = tm_players_data, aes(x = Assists)) +
  geom_histogram(bins = 30, fill = "blue", color = "black") +
  labs(title = "Histogram of Assists in Top 5 Leagues",
       x = "Assists", y = "Frequency"))
```

GOAT


Histogram of Assists in Top 5 Leagues
Frequency
200
150
100
50
0
0
100
200
300
Assists

GOAT

Top 5 players with the maximum number of assists.

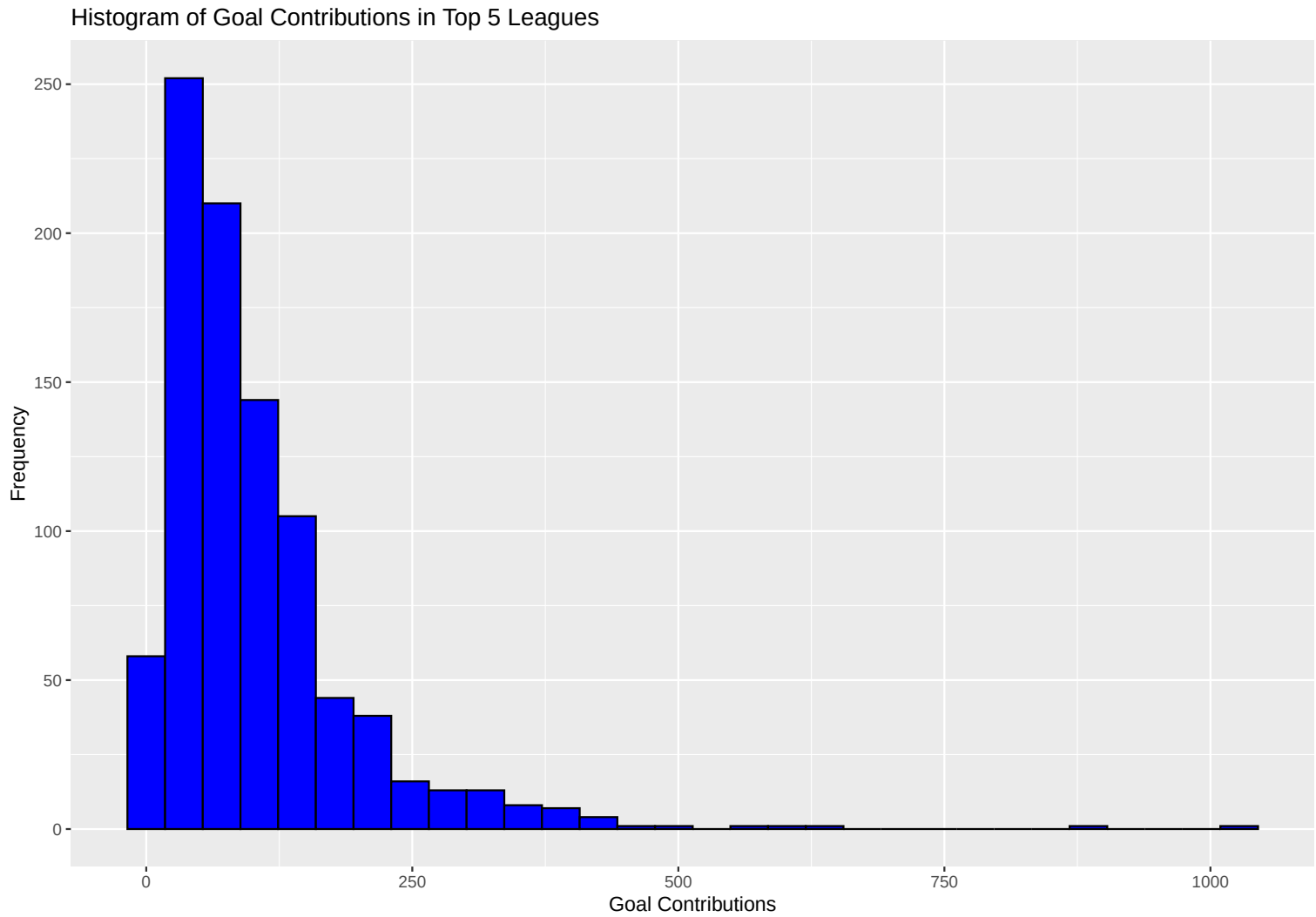
```
tm_players_data %>%
  arrange(desc(Assists)) %>%
    select(PlayerName,Assists) %>%
      head(5)
```

```
## # A tibble: 5 x 2
##   PlayerName        Assists
##   <chr>               <dbl>
## 1 Lionel Messi          334
## 2 Angel Di Maria        231
## 3 Kevin De Bruyne       213
## 4 Cristiano Ronaldo     212
## 5 Franck Ribery         210
```

GOAT

```
(goal_contributions_histogram <- ggplot(data = tm_players_data, aes(x = GoalContribu
  geom_histogram(bins = 30, fill = "blue", color = "black") +
  labs(title = "Histogram of Goal Contributions in Top 5 Leagues",
       x = "Goal Contributions", y = "Frequency"))
```

GOAT


Histogram of Goal Contributions in Top 5 Leagues
Frequency
250
200
150
100
50
0
0
250
500
750
1000
Goal Contributions

GOAT

Top 5 players with the maximum number of goal contributions.

```
tm_players_data %>%
  arrange(desc(GoalContributions)) %>%
    select(PlayerName,GoalContributions) %>%
      head(5)
```

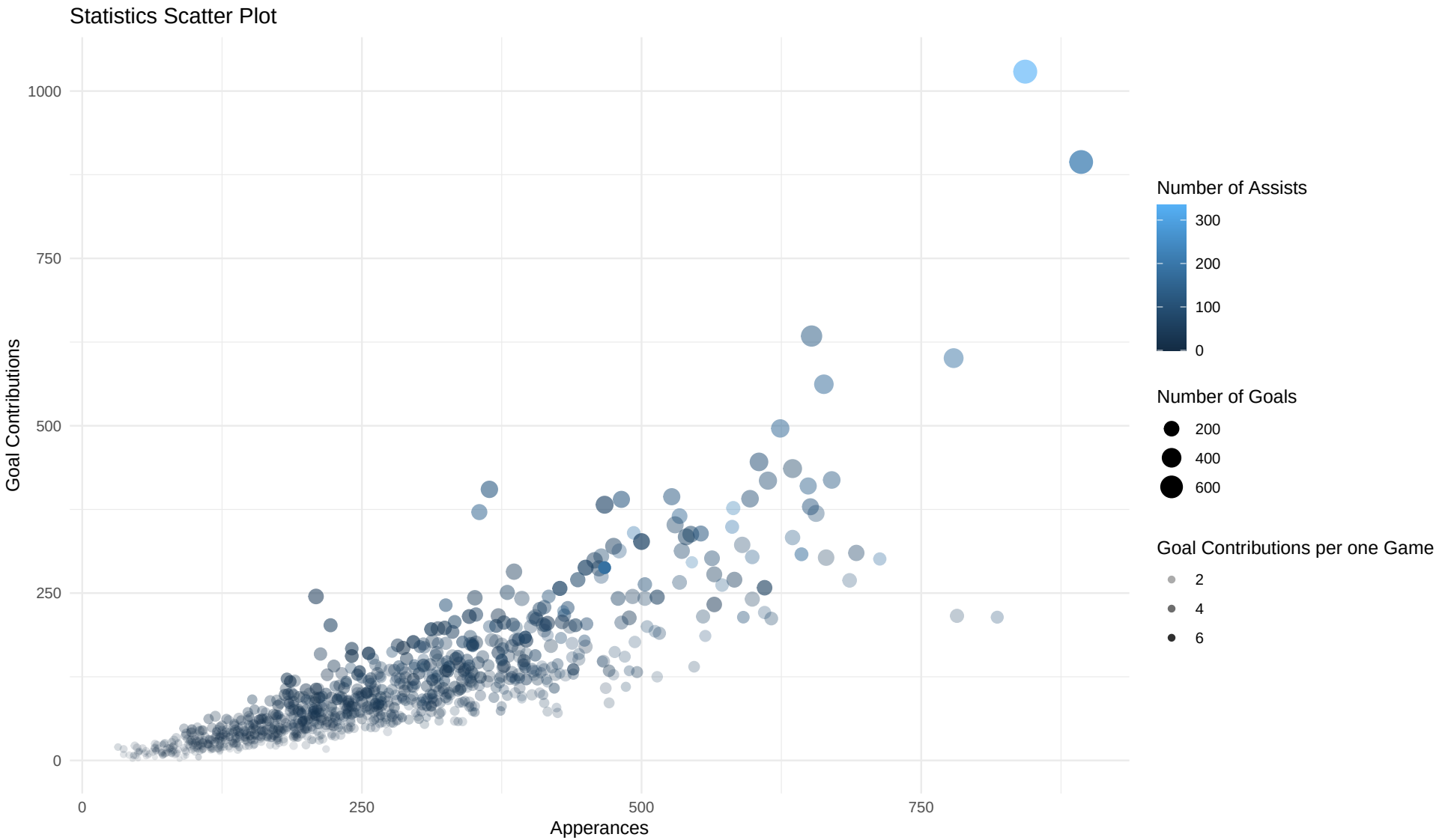
```
## # A tibble: 5 x 2
##   PlayerName         GoalContributions
##   <chr>                          <dbl>
## 1 Lionel Messi                    1029
## 2 Cristiano Ronaldo                894
## 3 Robert Lewandowski               634
## 4 Karim Benzema                    601
## 5 Zlatan Ibrahimovic               562
```

GOAT

Let's visualize all of the metrics in a scatter plot.

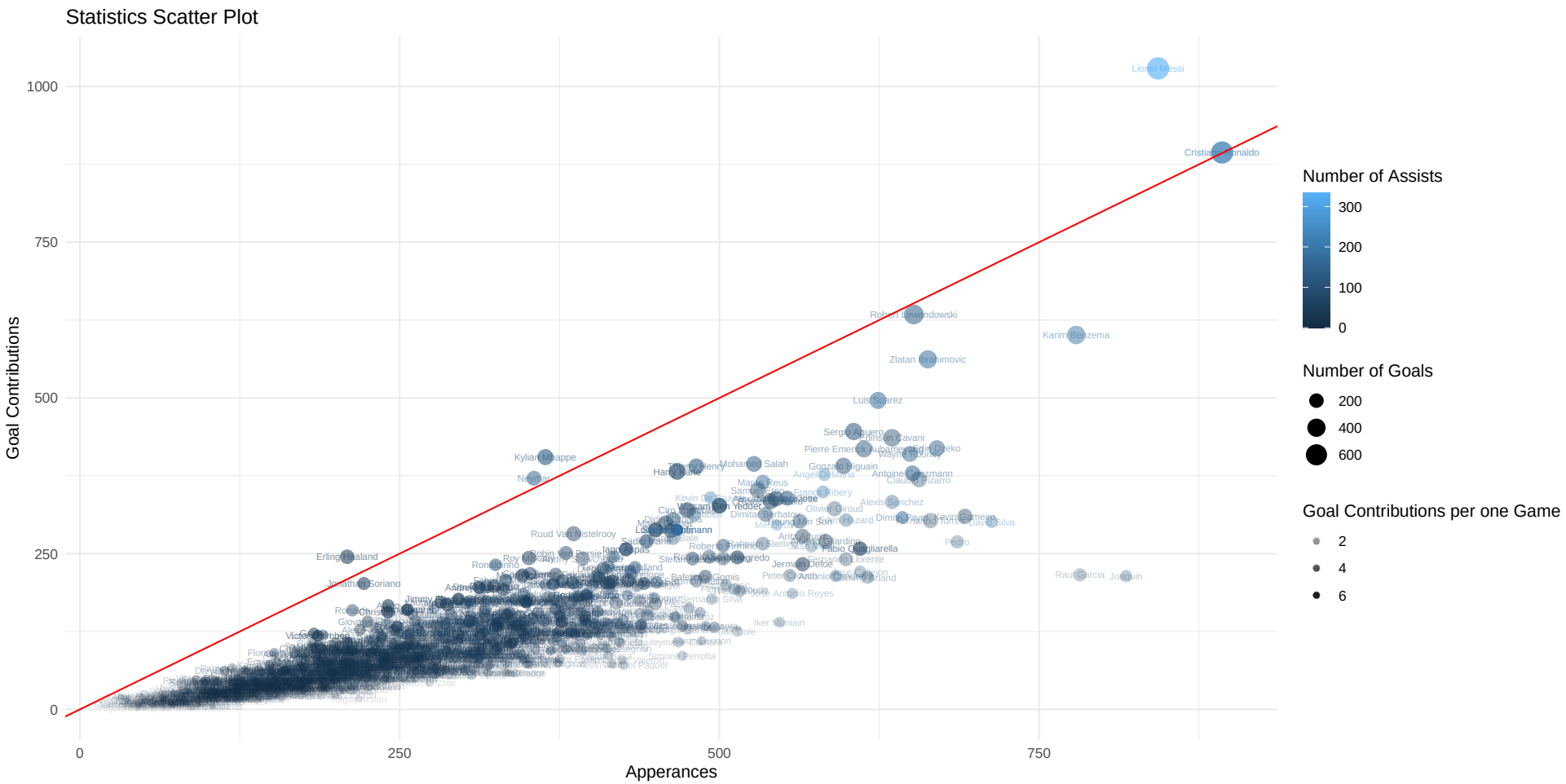
```
(stats_scatter_plot <- ggplot(data = tm_players_data,
    aes(x = Apperances, y = GoalContributions,
        size = Goals, color = Assists, alpha=GCPer90)) +
   geom_point() + labs(x = "Apperances", y = "Goal Contributions", title = "Statisti
        color = "Number of Assists", size = "Number of Goals", color = "Number of As
        ) + theme_minimal())
```

GOAT


Statistics Scatter Plot
Goal Contributions
1000
750
500
250
0
0
250
500
750
Apperances
Number of Assists
300
200
100
0
Number of Goals
200
400
600
Goal Contributions per one Game
2
4
6

GOAT

```
(stats_scatter_plot + geom_text(aes(label = PlayerName), size = 2) + geom_abline(intercept = 0, slope = 1, color = "red"))
```



GOAT

- What is one thing that is in common in all of the distributions we analyzed ?
- How to get an all in one metric for all of the players to evaluate the level of their greatness ?
- Which of these statistics were the hardest and most competitive to achieve ?

GOAT

- None of them resembles a normal distribution.
- Almost all of the distributions had a long right tail.
- This is a good sign for the “GOAT”ness of the players.
- This picture will be visible in any field that required talent, skills, opportunity and persistence.

GOAT

Why don't the distributions follow a Gaussian distribution ?

GOAT

The distributions are more likely to follow a lognormal distribution.

The main charachteristics of a lognormal distribution are:

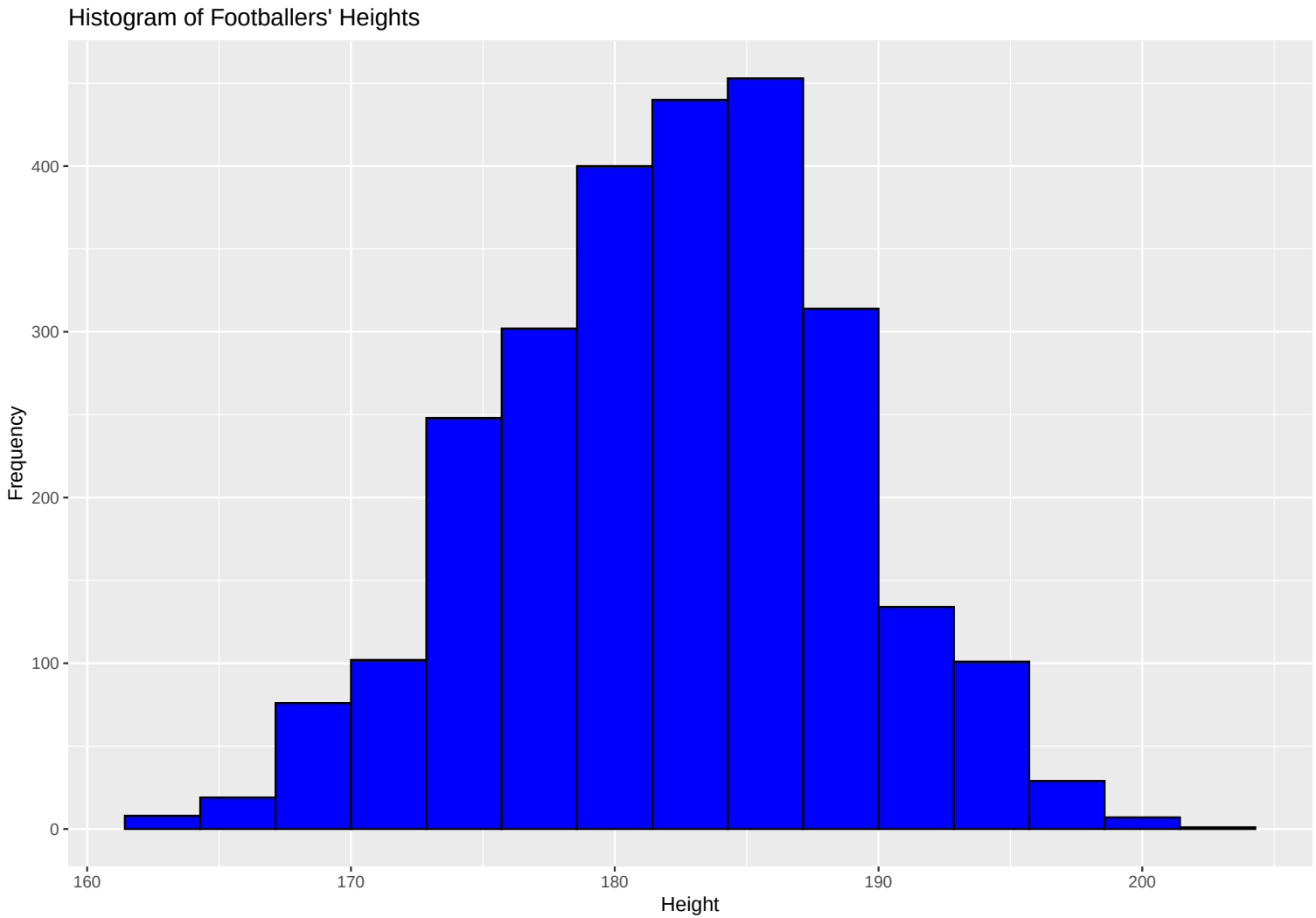
- The distribution is skewed to the right(Mean $>$ Median).
- The distribution is uni modal(has one clear peak).
- The logarithm of the data is normally distributed.
- The distributions of natural abilities are more likely to follow a normal distribution.
- The distributions of some abilities that require some work apart from nature are more likely to follow a log normal distribution.

GOAT

Let's analyze the height of the footballers

```
load("./data/players_height.rda")
(height_histogram <- ggplot(data = player_heights, aes(x = height)) +
  geom_histogram(bins = 15, fill = "blue", color = "black") +
  labs(title = "Histogram of Footballers' Heights",
       x = "Height", y = "Frequency"))
```

GOAT


Histogram of Footballers' Heights
Frequency
400
300
200
100
0
160
170
180
190
200
Height

GOAT

Let's analyze the log normality of the players' statistics.

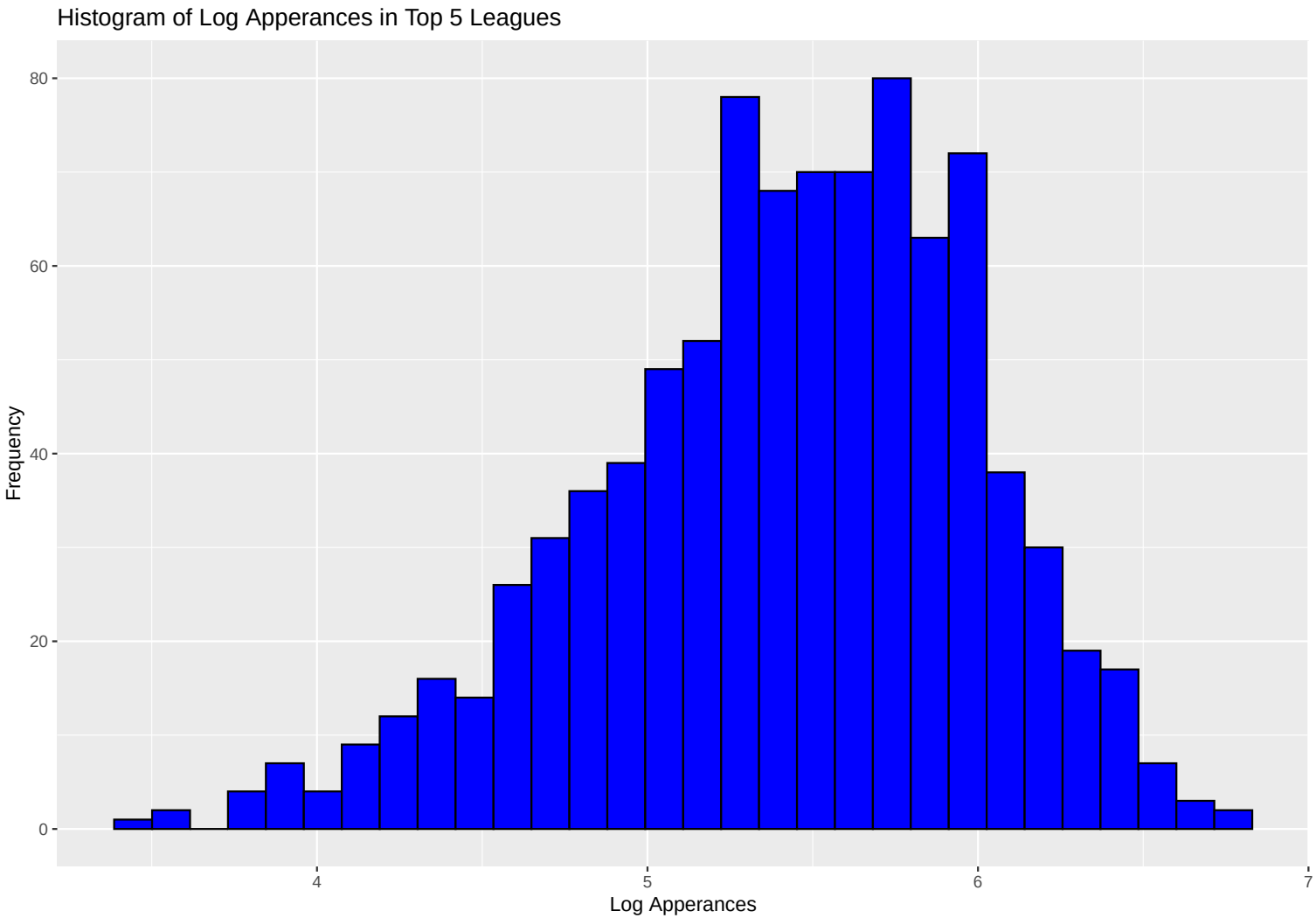
```
tm_players_data <- tm_players_data %>%
  mutate(log_Apperances = log(Apperances),
         log_Goals = log(Goals),
         log_Assists = log(Assists),
         log_GoalContributions = log(GoalContributions)
         )
```

GOAT

Let's visualize the lognormality of the data.

```
(log_apperances_histogram <- ggplot(data = tm_players_data, aes(x = log_Apperances))
  geom_histogram(bins = 30, fill = "blue", color = "black") +
  labs(title = "Histogram of Log Apperances in Top 5 Leagues",
       x = "Log Apperances", y = "Frequency"))
```

GOAT

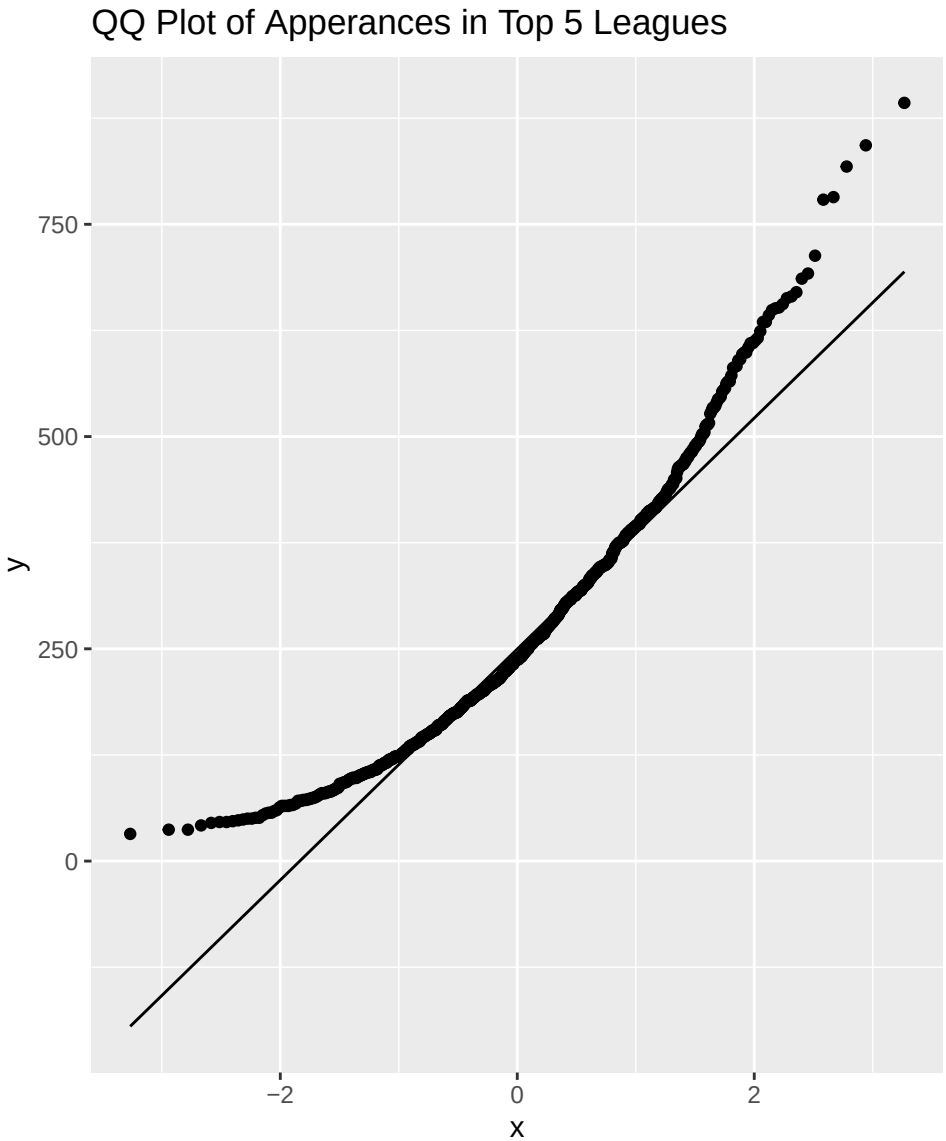

Histogram of Log Apperances in Top 5 Leagues
Frequency
80
60
40
20
0
4
5
6
7
Log Apperances

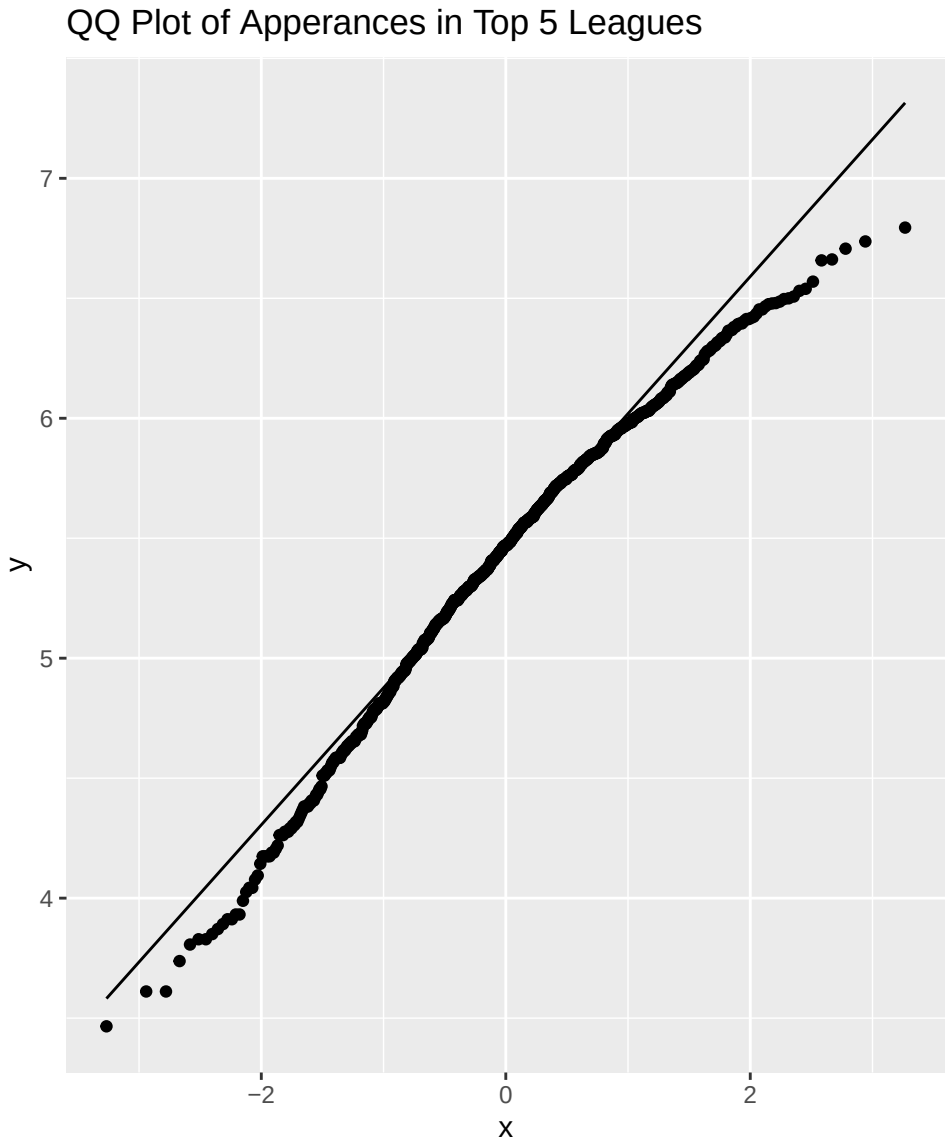
GOAT

Let's compare the qq plots.

```
(appereances_qq <- ggplot(data = tm_players_data, aes(sample = Apperances)) +
  stat_qq() +
  stat_qq_line() +
  labs(title = "QQ Plot of Apperances in Top 5 Leagues"))
(log_appearances_qq <- ggplot(data = tm_players_data, aes(sample = log_Apperances))
  stat_qq() +
  stat_qq_line() +
  labs(title = "QQ Plot of Log Apperances in Top 5 Leagues"))
grid.arrange(appereances_qq, log_appearances_qq, ncol = 2)
```

GOAT


QQ Plot of Apperances in Top 5 Leagues
750
500
250
0
y
−2
0
2
x

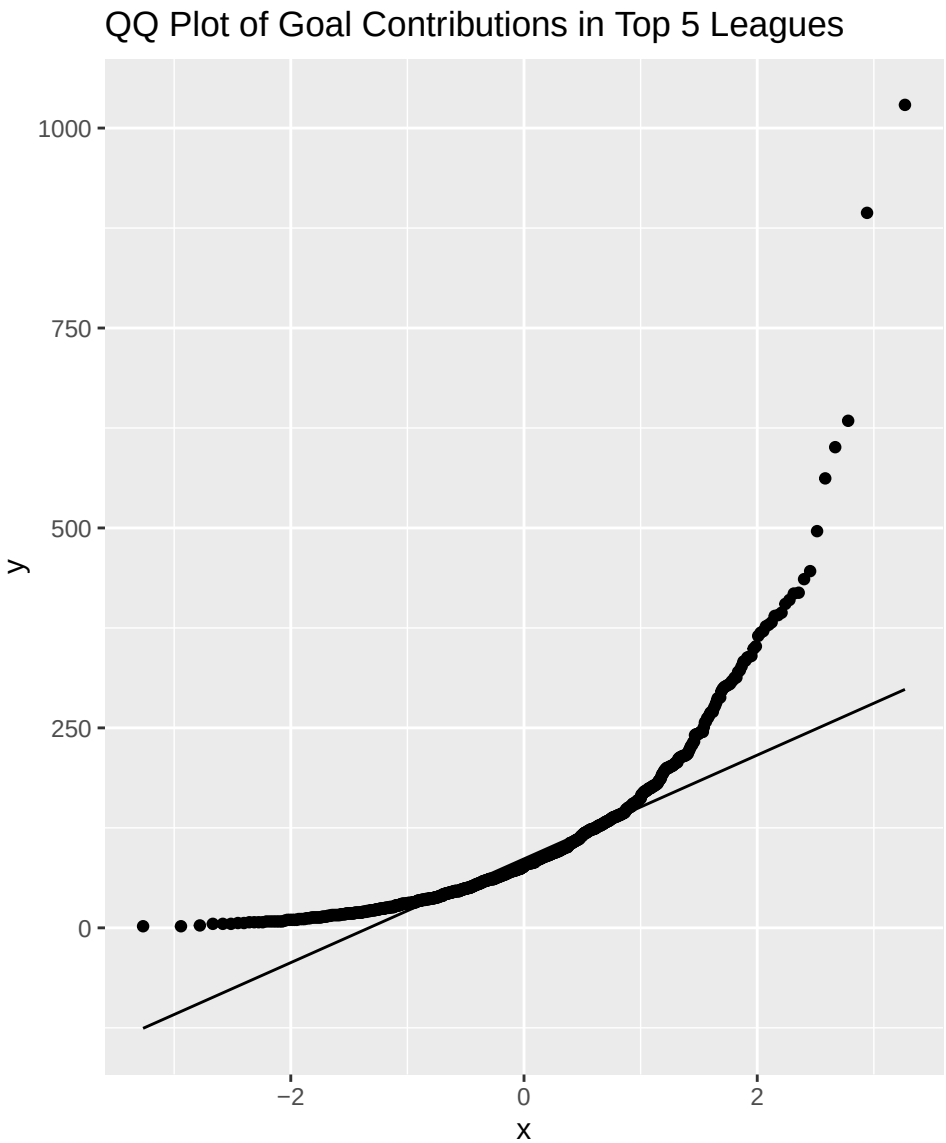

QQ Plot of Apperances in Top 5 Leagues
7
6
5
4
y
−2
0
2
x

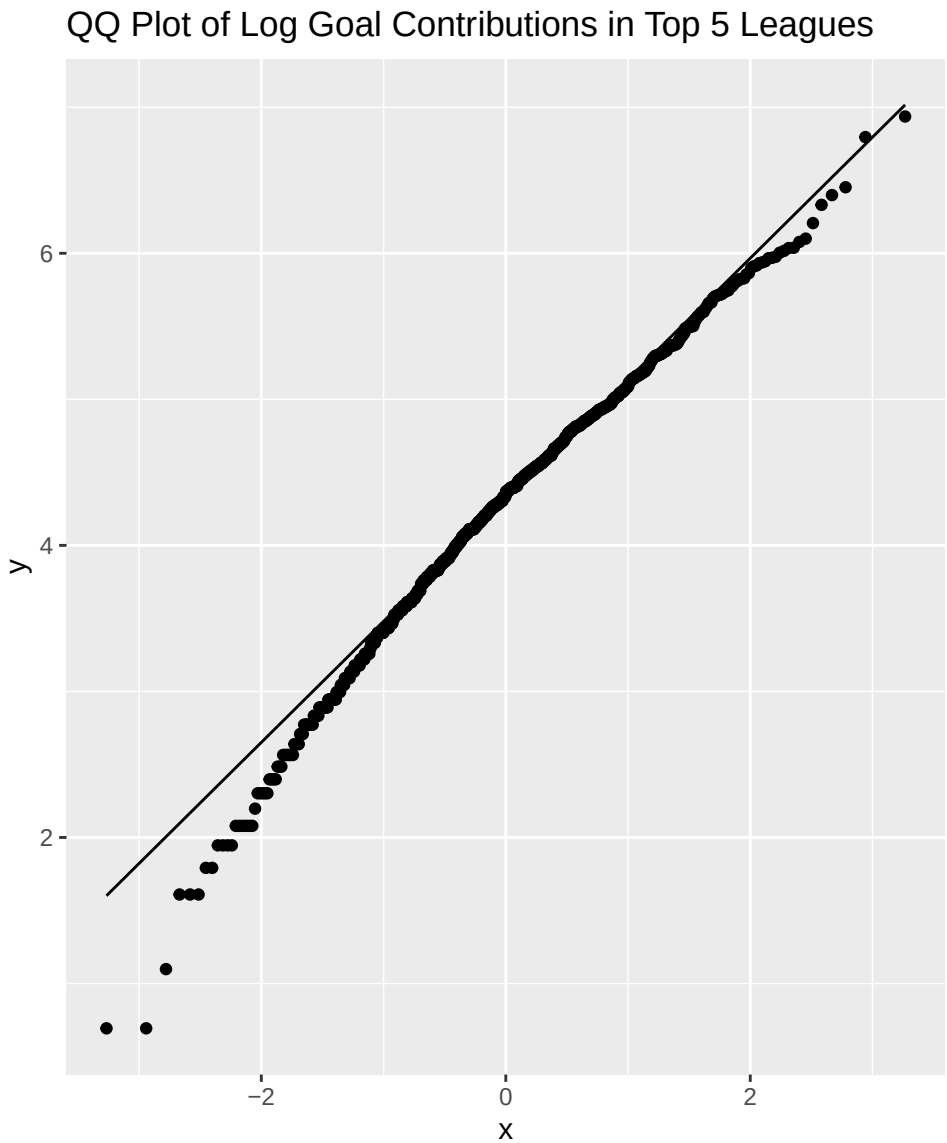
GOAT

Now the goal contributions

```
gc_qq <- ggplot(data = tm_players_data, aes(sample = GoalContributions)) +
  stat_qq() +
  stat_qq_line() +
  labs(title = "QQ Plot of Goal Contributions in Top 5 Leagues")
log_gc_qq <- ggplot(data = tm_players_data, aes(sample = log_GoalContributions)) +
  stat_qq() +
  stat_qq_line() +
  labs(title = "QQ Plot of Log Goal Contributions in Top 5 Leagues")
grid.arrange(gc_qq, log_gc_qq, ncol = 2)
```

GOAT


QQ Plot of Goal Contributions in Top 5 Leagues
1000
750
500
250
0
y
−2
0
2
x


QQ Plot of Log Goal Contributions in Top 5 Leagues
6
4
2
y
−2
0
2
x

GOAT

Why ?

Let's talk about two data generating processes.

- Proportional gain
- Weakest link

It is important to note that we are working with samples of data, so we can just say that they are well modeled by Gaussian or log normal distributions. Let's recall the Central Limit Theorem.

GOAT

The CENTRAL LIMIT THEOREM applies that if you add up some random factors, the sum will be *approximately* normally distributed.

- Mostly identical distributions.
- Mostly non correlated factors.
- Sample should be big enough to claim.

GOAT

Referring to the CENTRAL LIMIT COROLLARY, the product of multiple factors will tend to lognormal distribution, because multiplication of factors will be the same as adding the logarithgms.

$$Y = X * X => \log Y = \log X + \log X$$

GOAT

This limits help us to interpret why the performance metrics follow up a log normal distribution.

- They start with equal opportunities and abilities.
- In the end the distribution is different from normal.
- The difference in the distributions, explains the players' abilities to overcome an average player.

In other words, we are born GAUSSIAN, but we tend to become LOG-NORMAL. this implies to any field(finance, healthy lifestyle or else). So the best players' data is well simulated by proportional gain.

GOAT

Let's divide the factors affecting the greatness of the players into two categories.

Physical factors(Mostly well modelled by Gaussian):

- Height.
- Genetically inherited athletic abilities.

Other factors(well modeled by log normal):

- Opportunities for growth.
- Tactical understanding.
- Football IQ.

The elite players' distribution is modeled by the multiplication of all of these factors and even more.

GOAT

What about the weakest link process ?

Weakest link assumes that there is a risk factor for some ability, if not taken into account, they will face some limitations in their abilities.

- The players learn to obtain new abilities.
- Lack of attitude and resources will interfere the players from reaching world class.

GOAT

Achievement is lognormal.
Talent is Gaussian.

GOAT

What about the greatness score ?

How we can identify how far are the top players from their opponents ?

GOAT

Let's recall z-score.

- Z-score represents the number of standard deviations that a particular observation is away from the mean of the distribution.
- Z-score can be used to compare different distribution.

Z-score formula is:

$$Z_X = \frac{X - \mu_x}{\sigma_x}$$

GOAT

Let's calculate the average Z-score of the most important and competitive statistics for the player and identify how far did the best of the best get in that scale.

How we can identify the competitvenes of each statistic ?

GOAT

Let's recall the C5 and HHI indexes.

Concentration Index

$$C5 = \frac{\text{Number of (goal contribusions) earned by top N players}}{\text{Total number of (goal contribusions) earned by all players}}$$

GOAT

$$HHI = \sum_{i=1}^{n} w_i^2$$

Simple case: 2 firms in the market

- Highly competitive industry, both firms have the same market share

$HHI = \sum_{i=1}^{2} w_i^2 = 0.5^2 + 0.5^2$

- Monopoly, 1 firm has 100%

$HHI = \sum_{i=1}^{2} w_i^2 = 1^2 + 0^2$

GOAT

$$HHI = \sum_{i=1}^{n} w_i^2$$

lower and upper bounds

$$1/n \leq HHI \leq 1$$

GOAT

Several variations of the index exists.

$$HHI = \sum_{i=1}^{n} \left(\frac{gc_i}{\sum_{i=1}^{n} gc_i} \right)^2$$

gc_i is the number of the goal contributions the player has earned.

GOAT

Let's identify the most competitive metrics and calculate a weighted mean of z-scores to get a "greatness" score for the players.

GOAT

Calculating C5 metrics for the Minutes Played.

```
top5_minutes_played <- tm_players_data %>%
  arrange(desc(MinutesPlayed)) %>%
  select(MinutesPlayed) %>%
  head(5) %>%
  sum()
overall_minutes_played <- sum(tm_players_data$MinutesPlayed)
(c5_minutes_played <- (top5_minutes_played / overall_minutes_played)*100)
```

```
## [1] 1.707521
```

GOAT

Calculating C5 metrics for the Goals.

```
top5_goals <- tm_players_data %>%
  arrange(desc(Goals)) %>%
  select(Goals) %>%
  head(5) %>%
  sum()

overall_goals <- sum(tm_players_data$Goals)
(c5_goals <- (top5_goals / overall_goals)*100)
```

```
## [1] 4.467371
```

GOATs

Calculating C5 metrics for the Assists.

```
top5_assists <- tm_players_data %>%
  arrange(desc(Assists)) %>%
  select(Assists) %>%
  head(5) %>%
  sum()

overall_assists <- sum(tm_players_data$Assists)
(c5_assists <- (top5_assists / overall_assists)*100)
```

```
## [1] 3.644315
```

GOATs

Calculating C5 metrics for the number of games won, when the player was on the field.

```
top5_games_won <- tm_players_data %>%
  arrange(desc(GamesWon)) %>%
  select(GamesWon) %>%
  head(5) %>%
  sum()

overall_games_won <- sum(tm_players_data$GamesWon)
(c5_games_won <- (top5_games_won / overall_games_won)*100)
```

```
## [1] 2.285676
```

GOATs

Calculating C5 metrics for the Goal Contributions.

```
top5_gc <- tm_players_data %>%
  arrange(desc(GoalContributions)) %>%
  select(GoalContributions) %>%
  head(5) %>%
  sum()

overall_gc <- sum(tm_players_data$GoalContributions)
(c5_gc <- (top5_gc / overall_gc)*100)
```

```
## [1] 3.994845
```

GOAT

C5 index for the number of goal contributions per 90 minutes.

```
top5_gc_per90 <- tm_players_data %>%
  arrange(desc(GCPer90)) %>%
  select(GCPer90) %>%
  head(5) %>%
  sum()

overall_gc_per90 <- sum(tm_players_data$GCPer90)
(c5_gc_per90 <- (top5_gc_per90 / overall_gc_per90)*100)
```

```
## [1] 1.929316
```

GOAT

Now let' find the HHI index for the same metrics.

```
hhi_minutes_played <- sum((tm_players_data$MinutesPlayed /
                           sum(tm_players_data$MinutesPlayed))^2)
hhi_goals <- sum((tm_players_data$Goals /
                    sum(tm_players_data$Goals))^2)
hhi_assists <- sum((tm_players_data$Assists /
                      sum(tm_players_data$Assists))^2)
hhi_games_won <- sum((tm_players_data$GamesWon /
                      sum(tm_players_data$GamesWon))^2)
hhi_gc <- sum((tm_players_data$GoalContributions /
                 sum(tm_players_data$GoalContributions))^2)
hhi_gc_per90 <- sum((tm_players_data$GCPer90 /
                       sum(tm_players_data$GCPer90))^2)
```

GOAT

```
(hhi_minutes_played)
```

```
## [1] 0.001328782
```

```
(hhi_goals)
```

```
## [1] 0.002214332
```

```
(hhi_assists)
```

```
## [1] 0.0020549
```

```
(hhi_games_won)
```

```
## [1] 0.001504224
```

```
(hhi_gc)
```

```
## [1] 0.00201605
```

```
(hhi_gc_per90)
```

```
## [1] 0.001486876
```

GOAT

Now let's get the weight for each statistic, as a low score for HHI identifies a more competitive market, let's take the inverse for the weights.

```
goals_weight <- (c5_goals + 1/hhi_goals) / 2
minutes_weight <- (c5_minutes_played + 1/hhi_minutes_played) / 2
assists_weight <- (c5_assists + 1/hhi_assists) / 2
games_won_weight <- (c5_games_won + 1/hhi_games_won) / 2
gc_per90_weight <- (c5_gc_per90 + 1/hhi_gc_per90) / 2
```

GOAT

Now let's get the z score for all of these metrics

```
tm_players_data <- tm_players_data %>%
  mutate(z_minutes = (MinutesPlayed - mean(MinutesPlayed)) / sd(MinutesPlayed),
         z_goals = (Goals - mean(Goals)) / sd(Goals),
         z_assists = (Assists - mean(Assists)) / sd(Assists),
         z_games_won = (GamesWon - mean(GamesWon)) / sd(GamesWon),
         z_gc = (GoalContributions - mean(GoalContributions)) / sd(GoalContributions
         z_gc_per90 = (GCPer90 - mean(GCPer90)) / sd(GCPer90)
         )
```

GOAT

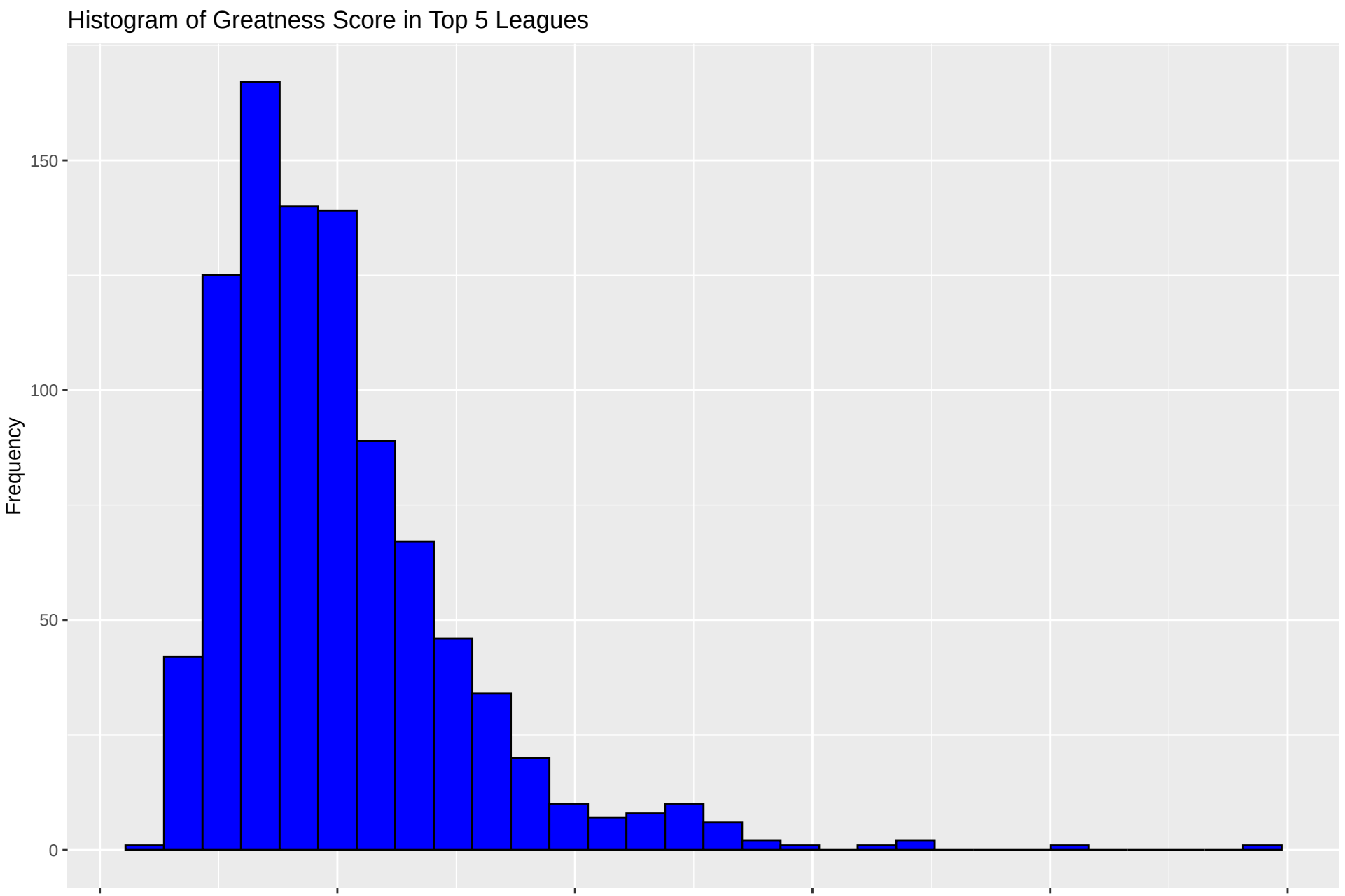
Calculating the weighted mean of the z-scores for each player.

```
tm_players_data <- tm_players_data %>%
  mutate(greatness_score = (z_minutes * minutes_weight +
                            z_goals * goals_weight +
                            z_assists * assists_weight +
                            z_games_won * games_won_weight +
                            z_gc_per90 * gc_per90_weight) / 5)
```

GOAT

```
(greatness_score_histogram <- ggplot(data = tm_players_data, aes(x = greatness_score
  geom_histogram(bins = 30, fill = "blue", color = "black") +
  labs(title = "Histogram of Greatness Score in Top 5 Leagues",
       x = "Greatness Score", y = "Frequency"))
```

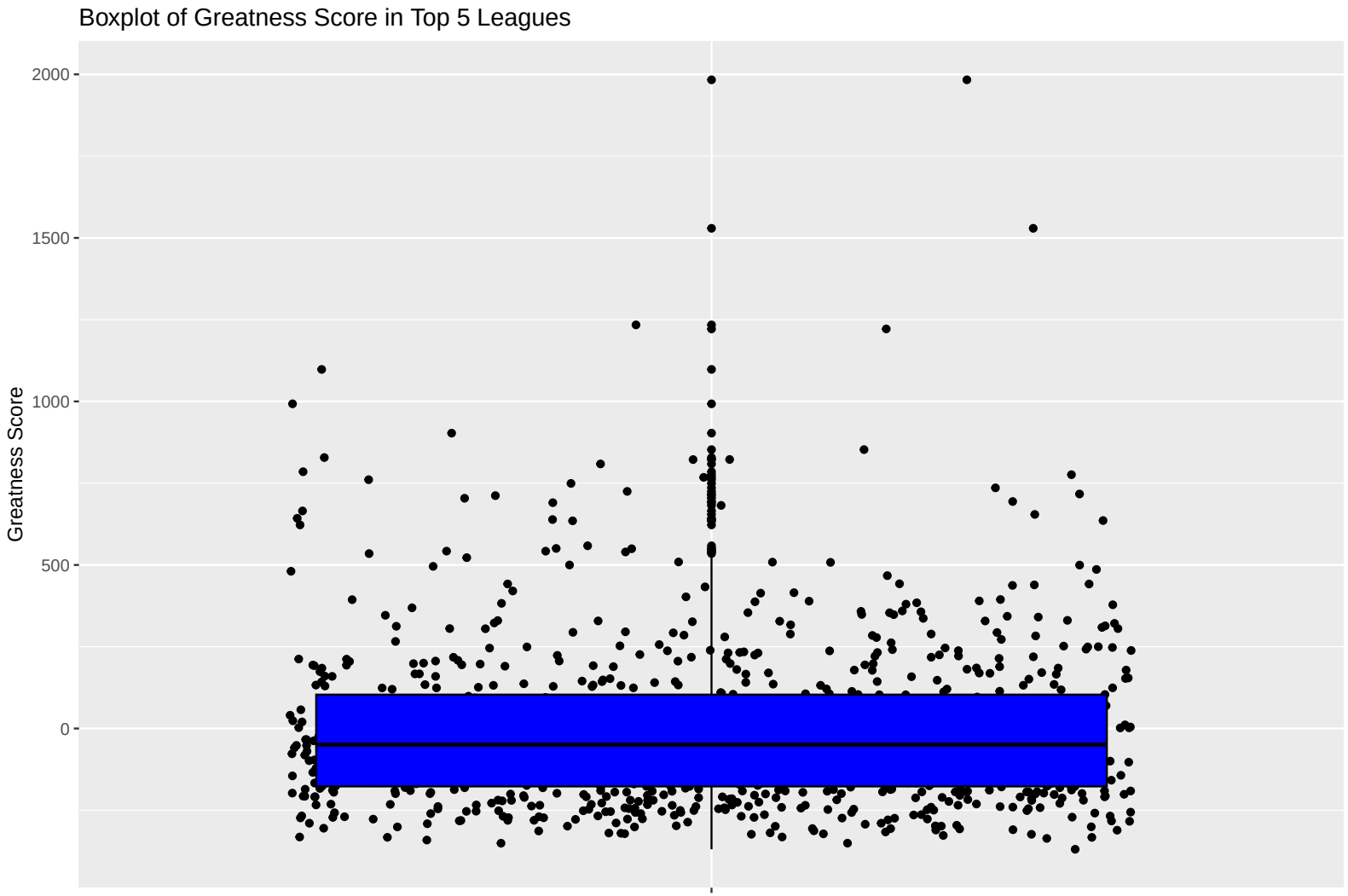
GOAT


Histogram of Greatness Score in Top 5 Leagues
Frequency
150
100
50
0

GOAT

```
(greatness_score_boxplot <- ggplot(data = tm_players_data, aes(x = "", y = greatness
  geom_boxplot(fill = "blue", color = "black") +
  labs(title = "Boxplot of Greatness Score in Top 5 Leagues",
       x = "", y = "Greatness Score"))
```

GOAT


Boxplot of Greatness Score in Top 5 Leagues
Greatness Score
2000
1500
1000
500
0

GOAT

Let's identify the top 10 players with the highest greatness score. Any big names missing ? What other factors need to be considered ?

```
tm_players_data %>%
  arrange(desc(greatness_score)) %>%
    select(PlayerName,greatness_score) %>%
      head(10)
```

```
## # A tibble: 10 x 2
##    PlayerName         greatness_score
##    <chr>                        <dbl>
##  1 Lionel Messi                 1983.
##  2 Cristiano Ronaldo            1529.
##  3 Karim Benzema                1234.
##  4 Robert Lewandowski           1222.
##  5 Zlatan Ibrahimovic           1098.
##  6 Angel Di Maria                993.
##  7 Luis Suarez                   903.
##  8 Edinson Cavani                853.
##  9 Franck Ribery                 828.
## 10 Mesut Ozil                    822.
```

GOAT

In game statistics are very crucial for determining the player's importance for the team. But most of the time the goals and assists make the headlines. The players who score bunch of goals are remembered for decades, but the midfielders or defenders who take care of the "dirty work" are not remembered as much.

Basketball


TORONTO RAPTORS
ATLANTIC DIVISION CHAMPIONS
2013-14
TORONTO RAPTORS
ATLANTIC DIVISION CHAMPIONS
2014-15
TORONTO RAPTORS
ATLANTIC DIVISION CHAMPIONS
2015-16
AIR CANADA CENTRE
TORONTO MAPLE LEAFS

Individual Performance - Basketball

$$EFF = PTS + AST + STL + BLK - MFG - MFT - TOV$$

- PTS : Points scored by the player.
- AST : Assists, when a player helps a teammate score.
- STL : Steals, when a player takes the ball from an opponent.
- BLK : Blocks, when a player prevents an opponent's shot.
- MFG : Missed Field Goals, a negative stat.
- MFT : Missed Free Throws.
- TOV : Turnovers, or lost possessions.

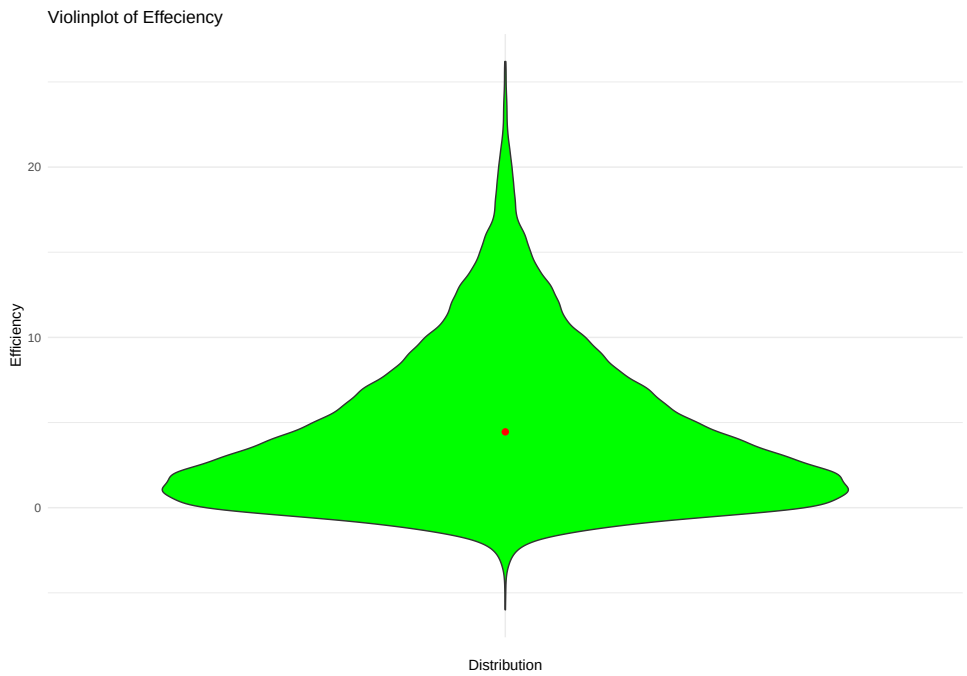
Efficiency

```
# EFF
nba = nba %>%
  mutate(
    MFG = FGA - FG,  #missed FG
    MFT = FTA - FT, #missed FT
    EFF = (PTS + AST + STL + BLK - MFG - MFT - TOV)
  )
```

Efficiency

```
ggplot(nba, aes(x = "", y = EFF))+
  geom_violin(fill = "green") +
  stat_summary(fun = mean, geom = "point", shape = 20, size = 3, color = "red")+
  labs(title="Violinplot of Effeciency",y="Efficiency",x="Distribution")+
  theme_minimal()
```

Efficiency



Efficiency

Table 1: TOP 10 Effeciency

Player	EFF
Nikola Jokic	26.2
Michael Jordan	26.0
Damian Lillard	26.0
Joel Embiid	25.8
Shai Gilgeous-Alexander	25.7
Damian Lillard	25.5
James Harden	25.0
Nikola Jokic	25.0
Kevin Durant	25.0
Luka Doncic	24.9

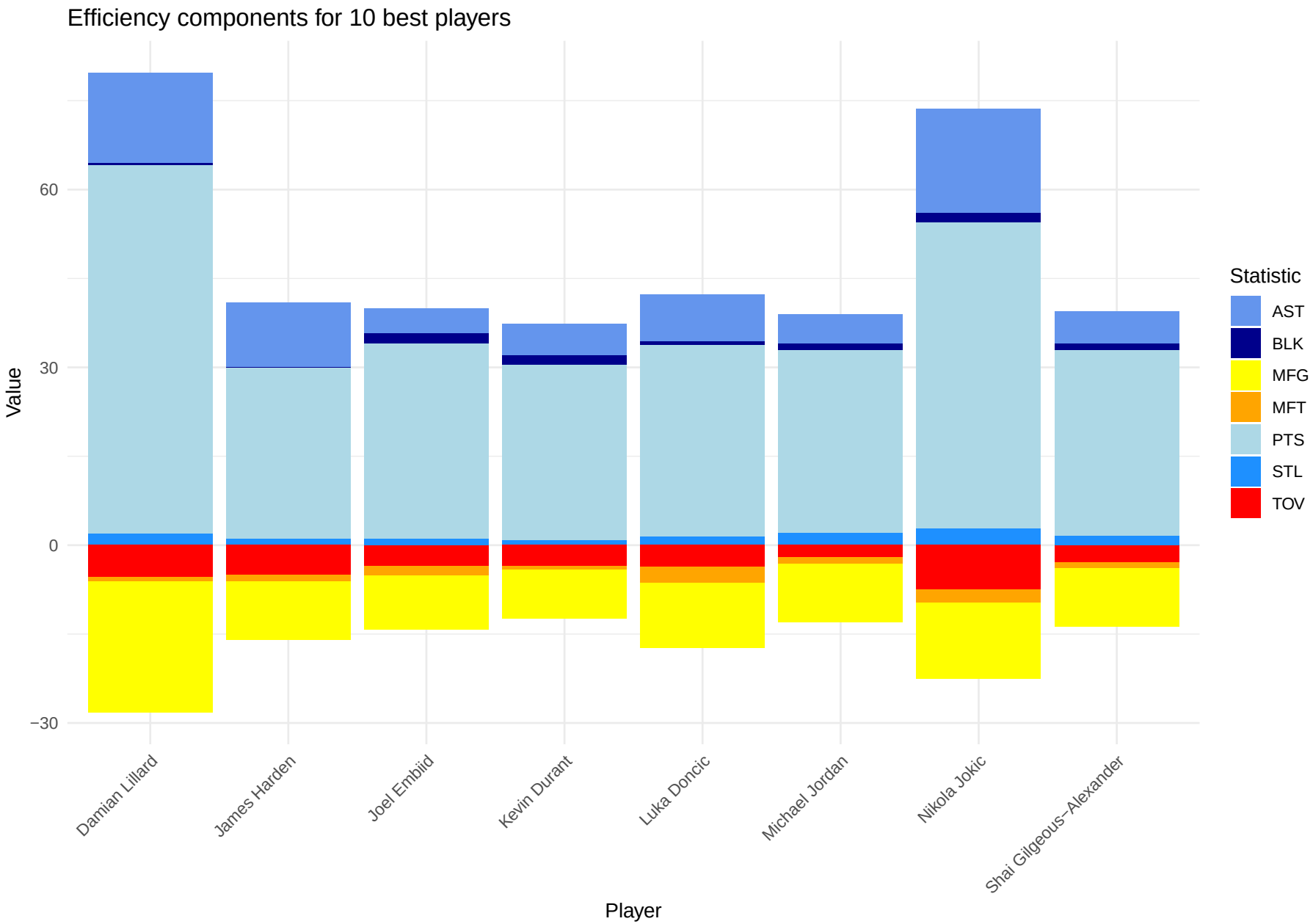
Efficiency

```
Top_EFF_graph = Top_EFF %>%
  select(Player, PTS, AST, STL, BLK, MFG, MFT, TOV) %>%
  mutate(MFG = -MFG,
         MFT = -MFT,
         TOV = -TOV)
colors <- c(
  "PTS" = "#ADD8E6",
  "AST" = "#6495ED",
  "STL" = "#1E90FF",
  "BLK" = "#00008B",
  "MFG" = "yellow",
  "MFT" = "orange",
  "TOV" = "red"
)
```

Efficiency

```
Top_EFF_graph = pivot_longer(Top_EFF_graph, cols = -Player, names_to = "Statistic",
ggplot(Top_EFF_graph, aes(x = Player, y = Value)) +
  geom_bar(stat = "identity", aes(fill = Statistic)) +
  scale_fill_manual(values = colors) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "Efficiency components for 10 best players")
```

Efficiency


Efficiency components for 10 best players
Value
60
30
0
−30
Damian Lillard
James Harden
Joel Embiid
Kevin Durant
Luka Doncic
Michael Jordan
Nikola Jokic
Shai Gilgeous−Alexander
Player
Statistic
AST
BLK
MFG
MFT
PTS
STL
TOV

Longevity

```
nba_app = nba %>%
  group_by(Player) %>%
  summarise(G = sum(G))
```

Longevity

```
#Top
nba_app_10 = nba_app %>% arrange(desc(G)) %>%
  head(10)
```

Longevity

```
knitr::kable(nba_app_10 %>% select(Player, G), caption = "TOP 10 Basketball")
```

Table 2: TOP 10 Basketball

Player	G
Vince Carter	1691
Andre Miller	1562
Jason Kidd	1526
Dirk Nowitzki	1522
Kevin Garnett	1509
Joe Johnson	1490
Kyle Korver	1444
LeBron James	1421
Gary Payton	1415
Juwan Howard	1412

Normalisation

$$N_{ij} = \frac{M_{ij}}{\overline{M}_i}$$

where:

- N_{ij} is the normalized number of games for player j in year i,
- M_{ij} is the number of games played by player j in year i,
- $\overline{M}_i$ is the average number of games played by all players in year i.

Normalisation

```
#Basket
nba_app_avg = nba %>%
  group_by(Year) %>%
  summarise(G_avg = mean(G, na.rm = TRUE))

nba = merge(nba, nba_app_avg, by = "Year")

nba_norm = nba %>%
  group_by(Player) %>%
  summarise(Norm = sum(G/G_avg))
```

Standardisation

$$S_{ij} = \frac{N_{ij}}{\max(M_i)} \times 100$$

- S_{ij} is the standardized number of games for player j in year j,
- N_{ij} is the normalized number of games for player j in year j,
- $\max(M_i)$ is the the maximum standardized number of games among all players in year i,

Standardisation

```r
#Basket
nba_norm$Str <- (nba_norm$Norm / max(nba_norm$Norm)) * 100
```

Standardisation

```
# Createa a histogram of STR

top10_longevity <- nba_norm %>%
  arrange(desc(Str)) %>%
  head(10)

knitr::kable(top10_longevity %>% select(Player, Str), caption = "TOP 10 Longevity in
```

Table 3: TOP 10 Longevity in Basketball

Player	Str
Vince Carter	100.00000
Andre Miller	89.09289
Dirk Nowitzki	89.06225
Jason Kidd	87.91445
Kevin Garnett	87.42957
LeBron James	85.95175
Joe Johnson	85.74336
Kyle Korver	85.69703
Jeff Green	85.07005
D.J. Augustin	81.20257